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Utilization of geopolymer in wood wool insulation boards: Design optimization, development and performance characteristics

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ABSTRACT

This study investigates the substitution of ordinary Portland cement (OPC) with geopolymer in the production of wood wool geopolymer boards (WWGB), offering insights into manufacturing methods, design parameters, and performance characteristics. The composite formulation involves fibre pre-treatment, adjustment of precursor components, and control of activator compositions. The optimized formulation results in a lightweight composite with a density of 392 kg m^{-3} and porosity of 76 %, meeting prescribed minimum compressive (20 kPa) and bending strength (1700 kPa) requirements. The inclusion of natural fibres influences the optimal Na_2O concentration and modulus for strength, with treated fibres recommended for higher modulus applications and improved strengths. Furthermore, it exhibits low thermal conductivity at 77 mW m^{-1} K^{-1}, minimal moisture sorption, and low vapour resistance, offering a sustainable alternative in the field of insulation materials.

1. Introduction

Wood wool cement board (WWCB), comprising wood wool fibres bonded with ordinary Portland cement (OPC), has gained widespread usage in construction due to its resistance to decay and insects, lightweight nature, and favourable acoustic and thermal insulation properties (Kochova, 2022; Van Elten, 2006). However, OPC manufacturing has a high embodied carbon, resulting in approximately 600 to 850 kg of CO_2 emissions per tonne (Nature, 2021; Feiz et al., 2015). Sustainable alternatives are sought, with geopolymers emerging as promising candidates that offer significant carbon footprint reduction (Duxson et al., 2007) and potential performance advantages over OPC (Provis and Bernal, 2014). Exploration of wood wool geopolymer board (WWGB) could potentially address the environmental concerns; however, wood-cement compatibility, optimal binder formulation, fibre conditions, and manufacturing methods need clarification.

A key challenge in manufacturing wood-cement composites arises from the chemical incompatibility between wood and cement (Na et al., 2014; Fan et al., 2012). Saccharides in natural fibres can inhibit cement setting and hardening when leached out (Kochova et al., 2017; Grée et al., 2017). However, promising compatibility has been reported between geopolymer binders and wood fibres (Tan et al., 2019; Furtos et al., 2021, 2019; Ye et al., 2023), as well as other natural fibres (Kuqo et al., 2023; Liu et al., 2019; Sankar et al., 2017; Ribeiro et al., 2016; Assaedi et al., 2015; Alzeer and MacKenzie, 2013; Alves et al., 2019; Chen et al., 2013). Substituting OPC with geopolymer holds promise in producing WWGB.

Tailoring the geopolymer binder is crucial in determining WWGB performance. Geopolymers, formed by activating industrial aluminosilicate waste materials like coal fly ash (FA) and ground granulated blast furnace slag (GGBFS) using alkali agents (Provis and Bernal, 2014; Çelikten, 2021), have been extensively studies. The geopolymerisation of ASTM Class F FA, offering a low-calcium alkali-activated system with the primary reaction product of sodium aluminosilicate hydrate gel (N-A-S-H), provides better thermal stability compared to OPC or other alkali-activated materials (Provis and Bernal, 2014). The formulation of alkali activators, with varying SiO_2-to-Na_2O ratios (modulus) and Na_2O concentrations, significantly influences geopolymer gel formation. Furthermore, the inclusion of GGBFS in the FA-based precursor can enhance reactivity, leading to quicker setting and early strength development (Nath and Sarker, 2014). Therefore, investigating geopolymer formulation for WWGB production is essential.

Effective adhesion of fibres to geopolymer gels and potential surface morphology enhancement through pre-treatment are vital. Alkaline treatment increases fibre hydrophobicity and roughness (Wimmer et al., 2023), potentially improving adhesion. Alkali-treated wood fibres have demonstrated improved mechanical performance within cementitious matrices (Tuğluca et al., 2023), and efforts have been made to use

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alkali-treated fibres for geopolymer reinforcement (Chen et al., 2013; Malenab et al., 2017; Amalia et al., 2017). Therefore, WWGB made from pre-treated fibres is expected to exhibit improved adhesion and structural interlocking between fibres and the geopolymer binder.

Maximizing fibre content is critical for achieving the desired insulation properties. However, most studies have focused on using fibres as reinforcing components in geopolymer composites (Kuqo et al., 2023; Liu et al., 2019; Tan et al., 2019; Sankar et al., 2017; Ribeiro et al., 2016; Assaedi et al., 2015; Alzeer and MacKenzie, 2013; Alves et al., 2019; Chen et al., 2013; Grée et al., 2019). For instance, Kuqo et al. (2023) examined the dry mixing-spraying process to produce geopolymer-bonded fibreboards with up to 50 wt.% fibre content. Although the composite exhibited good bending strength, bulk densities exceeded 1000 kg m^{-3} and no thermal conductivity measurements were reported. Liu et al. (2019) investigated the production of wheat straw geopolymer composites, and the resulting boards exhibited relatively low densities ranging from 235 to 894 kg m^{-3} and thermal conductivities ranging from 92 to 186 mW m^{-1} K^{-1}. However, these values were less favourable compared to commercially available WWCB products, which have thermal conductivities of 80 to 90 mW m^{-1} K^{-1} with densities of 390 to 420 kg m^{-3} (MASEA, 2023). Therefore, manufacturing methods should be further explored to close the performance gaps.

This study aims to replace the high embodied carbon raw material (OPC) with geopolymer in the production of wood wool boards. Specifically, we will focus on the manufacturing method, the effect of spruce wood wool with and without alkali pre-treatment for binding with geopolymer, the inclusion of GGBFS into the FA-based precursor, and the modulus and Na_2O concentration of the alkali activator to optimize geopolymer gel formation for WWGB. By investigating the feasibility, characteristics and performances of a wood wool board based on geopolymer, this study aims to address the increasing demand for environmentally friendly alternatives in the field of insulation materials.

2. Material and methods

2.1. Raw materials

Locally sourced raw materials, namely spruce wood wool (WW), Class F fly ash (FA) and ground granulated blast furnace slag (GGBFS) obtained from the Netherlands, are utilized. The average particle size (d_{50}) of FA and GGBFS is 13.9 μm and 19.3 μm, respectively, as determined by a laser particle size analyzer (Mastersizer, 2000). The WW, with dimensions of 1 mm width, 0.3 mm thickness, and 25 cm length, is stored at 50 ± 5 % relative humidity and 20 ± 2 °C. The chemical composition of FA and GGBFS is analysed using X-ray fluorescence spectrometry (XRF) (PANalytical Epsilon 3), and their loss on ignition (LOI) is determined between 105 and 1000 °C, as presented in Table 1. For the alkali activator, sodium silicate solution (27.7 % SiO_2, 8.4 % Na_2O and 63.9 % H_2O) and sodium hydroxide (NaOH) pellets are employed. Furthermore, 1 wt.% sodium hydroxide solution is prepared for fibre pre-treatment.

Table 1
Chemical composition of FA and GGBFS.

Oxides (wt.%)	FA	GGBFS
SiO_2	54.57	29.41
Al_2O_3	21.60	13.21
Fe_2O_3	9.04	0.37
CaO	6.12	41.67
K_2O	2.85	0.42
MgO	1.17	8.57
SO_3	0.41	2.64
Other	2.13	1.07
LOI (1000 °C)	2.11	1.15
Specific density (kg m^{-3})	2225	2760

2.2. Preparation of samples

2.2.1. Boards design

Life cycle assessments comparing the replacement of OPC with geopolymer (Ouellet-Plamondon and Habert, 2015; Teh et al., 2017; Turner and Collins, 2013; Komkova and Habert, 2023; Davidovits, 2015) generally concur that geopolymer production yields a lower CO_2 footprint than OPC. However, the extent of reduction in global warming potential (GWP) varies based on assessment methods, assumptions, and data quality. Notably, alkali activators, namely sodium hydroxide with a GWP of 1.27 kg CO_2 kg^{-1} (Teh et al., 2017) and sodium silicate solution with a GWP of 0.424 kg CO_2 kg^{-1} (Fawer et al., 1999) emerge as major contributors to GWP. Higher Na_2O concentration and SiO_2-to-Na_2O modulus result in increased raw material consumption and, consequently, higher GWP. Striking a balance between environmental impact and overall performance is crucial when determining alkali activator design parameters. Additionally, both precursor materials (FA and GGBFS) are by-products of industrial processes and are subjected to emission allocation between primary product and by-product (Chen et al., 2010). Although studies suggest a potential higher GWP for GGBFS compared to FA (Flower and Sanjayan, 2007; Chen et al., 2010; Ouellet-Plamondon and Habert, 2015), the FA-to-GGBFS ratio is considered less critical for the overall environmental impact of the board.

In this study, four key parameters are investigated to achieve optimum performance of WWGB: (i) alkali pre-treatment of WW, (ii) Na_2O concentration, (iii) SiO_2-to-Na_2O modulus of the alkali activators, and (iv) percentage of GGBFS and FA in the precursor. A baseline geopolymer formulation is established with a blend weight ratio of 95 % FA and 5 % GGBFS for the precursor, along with an alkali activator with a modulus of 1.8 and Na_2O concentration of 8 % determined as optimal based on previous research (Luo et al., 2022a,b) and preliminary experiments.

From this baseline, the design parameters are varied, exploring different Na_2O concentrations ranging from 6 % to 12 %, modulus values spanning from 1.0 to 2.2, and different percentages of GGBFS ranging from 0 % to 15 %. Each geopolymer formulation is applied to both alkali pre-treated WW and untreated WW samples. The detailed compositions of the mixtures can be found in Fig. 1a.

To ensure optimal activation and workability during the mixing process with WW, the water-to-dry geopolymer (dry sodium silicate, sodium hydroxide, FA and GGBFS) ratio is set at 0.4. Furthermore, the weight percentage of WW in the total dry weight (WW and dry geopolymer) is set at 30 %. This percentage strikes a balance between effective binding and preventing binder runoff during the board-forming process. The density of WW without geopolymer is maintained at 90 kg m^{-3} by controlling the amount of WW added, ensuring consistency in the final volume. To ensure a homogenous coating of geopolymer pastes onto the WW within the designated fabrication timeframe, WW is pre-wetted by spraying with water and the water-to-fibres ratio is set at 0.4.

Energy expended in board fabrication, especially during the curing and heating process, is optimized. Preliminarily investigations consider both ambient and heat curing (Nath and Sarker, 2014; Kovalchuk et al., 2007; Sajan et al., 2021), with a decision in favour of heat curing at 60 °C for 24 h to achieve the required early strength.

2.2.2. Boards manufacturing

The production procedure is illustrated in Fig. 1b. WW is divided into two batches: one undergoing pre-treatment and the other remaining untreated. For the treated WW batch, immersion in a 1 wt.% sodium hydroxide solution is conducted for 24 h (Chandrasekar et al., 2017). Subsequently, the treated WW is rinsed with water to eliminate alkali residue and impurities and left to dry under ambient conditions. The alkali activators are synthesized beforehand by blending sodium silicate solution, sodium hydroxide pellets, and additional water based on predetermined formulations.

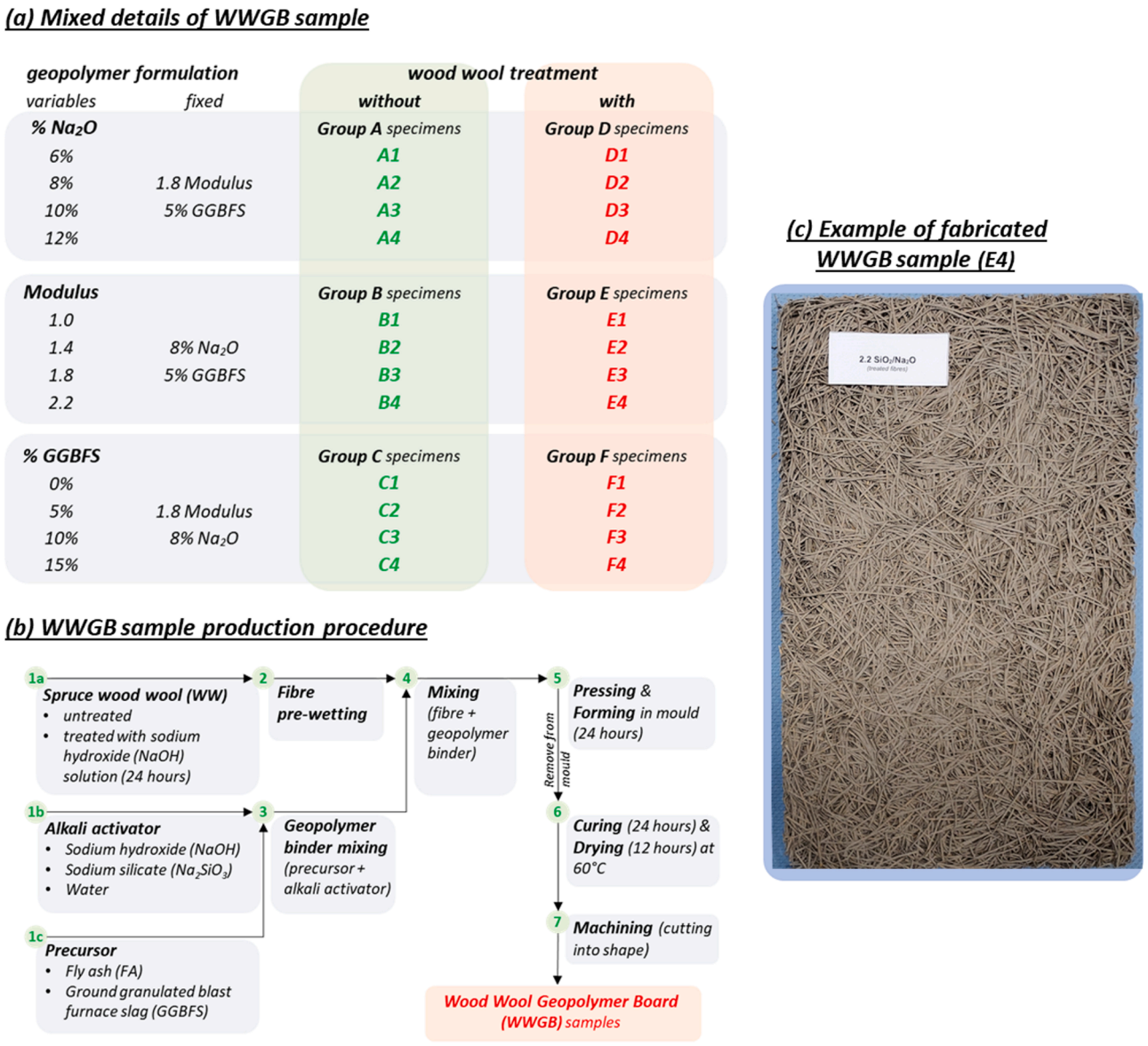


Fig. 1. (a) Mixed details for WWGB production, (b) WWGB sample production procedure, and (c) example of fabricated WWGB sample (E4).

During the main production process, WW is pre-wetted by spraying with water, using a water-to-fibre ratio of 0.4. The precursor materials (FA and GGBFS) and alkali activator solution are then added to a mixer and blended until a homogeneous paste is achieved. This geopolymer mixture is poured onto the WW and thoroughly mixed to ensure optimal coating of the binder onto the WW surfaces. The resulting mixture is transferred into a fixed-volume mould (350 × 250 × 15 mm^3) and compressed overnight. Throughout the forming process, the coated fibres are randomly laid and deagglomerated to ensure a consistent and homogeneous mixture.

After 24 h, the compressed board is removed from the mould, wrapped in plastic foil, and cured overnight in an oven at 60 °C for 24 h. Following curing, the plastic is removed, and the board is further dried in an oven at 60 °C for another 12 h. Subsequently, the board is taken out of the oven, representing the final WWGB, and trimmed to its final dimensions (300 × 200 × 15 mm^3). A total of 24 WWGB samples are fabricated in this study, encompassing 20 distinct recipes (Fig. 1c).

2.3. Characterization methods

2.3.1. Mechanical strength

The mechanical properties of the samples are evaluated using a mechanical testing system (Instron 5967) equipped with a 5 kN load cell. The evaluation is based on the minimum criteria outlined in the European Standard EN_13168 (CEN, 2015), which serves as the benchmark for this testing. EN_13168 establishes the requirements for factory-manufactured wood wool products, presented in the form of boards or slabs, designed for application in the thermal insulation of buildings.

For the measurement of bending strength, the testing method specified in EN_12089 (CEN, 2013a) is employed. Two test specimens per sample type are cut into dimensions of 150 × 125 × 15 mm^3. The test machine is set to a constant speed of 10 mm min^{-1}. The maximum force exerted during the bending test is used to calculate the bending strength (σ_b). The minimum required σ_b value of 1700 kPa (CEN, 2015) is used as a reference.

To determine the compressive strength at 10 % deformation (σ_{10}), the testing procedure outlined in EN_826 (CEN, 2013b) is followed. Two test specimens per sample type with dimensions of 50 × 50 × 15 mm^3

are cut. The specimens are compressed at a constant displacement rate of 1.5 mm min^{-1}. The compression stress at a strain of 10 % (σ_{10}) is recorded. The minimum required σ_{10} at 20 kPa (CEN, 2015) is used as a reference.

2.3.2. Insulation properties

The measurement of thermal conductivity (λ) is carried out using the transient line source method with a thermal needle probe (AP Isomet model 2104). The thermal needle probe, known for its accuracy of 5 % of the reading plus 0.001 W m^{-1} K^{-1} is employed. All samples are conditioned at a relative humidity of 50 ± 5 %, and the measurements are conducted at 20 ± 2 °C. Three measurements are taken at different locations on each board.

The hygroscopic sorption properties of the samples are determined following standard ISO_12571 (ISO, 2013). The moisture contents (w) of all samples are measured after conditioning them at 80 % relative humidity in a climate chamber (Memmert ICH750). However, for the sorption isotherms measurement from 0 % to 85 % relative humidity, only samples A2 and D2, which represent WWGB made of untreated and treated WW, are included. To monitor the moisture uptake, the samples are weighed at 24-h intervals using a digital balance. Constant mass is considered reached when three successive weighings show a mass loss change of less than 0.1 %.

The water vapour diffusion resistance factor (μ) is measured using the cup methods as outlined in standard ISO_12572 (ISO, 2016). Samples A2 and D2 are used to investigate the water vapour transmission properties. The measurement is considered complete when five successive readings of the water vapour transmission rate density vary within ±5 %.

The measurement of sound absorption coefficient (α) and sound transmission loss (TL) is conducted using two impedance tubes: BSWA SW422 and SW477. These impedance tubes cover different frequency ranges, with SW422 covering the range of 63–1600 Hz and SW477 covering the range of 800–6300 Hz. The SW422 requires samples with a diameter of 100 mm, while the SW422 requires samples with a diameter of 30 mm. Only samples A2 and D2 are included. The sound absorption coefficient is measured following ISO_10534-2 (ISO, 1998), while the sound transmission loss is based on the transfer function method.

2.3.3. Physical properties

Visual inspection is conducted using a scanning electron microscope (SEM) (Phenom ProX). Additionally, the cross-section of the samples is analysed using micro-computed tomography (micro-CT) analyses (Scanco μCT 100).

To determine the particle density ($\rho_{particle}$), a helium pycnometer (Micromeritics AccuPyc II 1340) with a 10 cm^3 cup is used. From the measured particle density, the porosity φ (%) of the specimen is calculated using Eq. (1)

$$\varphi = 1 - \frac{\rho_{bulk}}{\rho_{particle}} \quad (1)$$

where ρ_{bulk} represents the bulk density of the sample. The pycnometer has an accuracy of 0.06 %.

2.3.4. Chemical composition

The chemical composition of the samples is analysed using Fourier transform-infrared (FT-IR) spectroscopy in conjunction with an attenuated total reflection (ATR) attachment (PerkinElmer Frontier FT-IR). The FT-IR spectra are collected over a wavenumber range of 4000 to 400 cm^{-1}, with a resolution of 1 cm^{-1}.

Thermogravimetric analysis (TGA) is performed using a thermogravimetric analyser (TA Instruments TGA Q500). The heating process starts from room temperature and continues up to 900 °C, with a heating rate of 10 °C min^{-1}. The samples are analysed under a controlled nitrogen atmosphere with a constant flow rate of 60 ml min^{-1}.

3. Results and discussions

3.1. Mechanical strength

Fig. 2a illustrates the relationship between the bending strength of WWGB and various design parameters. Notably, the most significant increase in bending strength is observed with higher Na_2O concentration, followed by an increase in modulus and GGBFS percentage. However, when considering the minimum bending strength requirement specified in EN_13168 for a 15 mm thickness board (1700 kPa) (CEN, 2015), WWGB samples made with a minimum of 8 wt.% Na_2O concentration, a modulus of 1.8 or higher, and a GGBFS ratio above 5 % meet the minimum strength requirement.

All samples meet the minimum requirement specified in EN_13168 (CEN, 2015), which mandates a minimum compressive stress at 10 % strain (σ_{10}) of 20 kPa (CS(10/Y)10) for handling purposes. Notably, samples with higher Na_2O concentrations (samples A4 and E4) exhibit the highest compressive strengths. This observation indicates an increased formation of aluminosilicate products under higher Na_2O concentrations, resulting in a stronger binding mechanism. Moreover, it is evident that the compressive strength of WWGB samples reaches a plateau within the range of 1.4 and 2.2 modulus and with a GGBFS range of 10 % and 15 %.

From previous research (Sun et al., 2022), the optimal Na_2O concentration is found to be between 5.5 to 10 wt.% for a standalone geopolymer paste. An excessive Na_2O concentrations further hinders the subsequent formation of geopolymer gel (Luo et al., 2023). However, a higher Na_2O concentration at 12 wt.% is found to further increase the compressive strength in samples A4 and E4. This suggests that part of the excess hydroxide from the activator reacts with the hydroxyl groups present in the WW fibres (Chandrasekar et al., 2017), instead of inducing early gel precipitation (Sun et al., 2022). The presence of natural fibres significantly influences the optimal value of Na_2O concentration of an activator.

Compressive strength continues to increase up to a modulus of 1.8 for samples with untreated WW and 2.2 with treated WW in this study, under a fixed 8 wt.% Na_2O concentration. Notably, the compressive strength of sample B4 drops significantly; however, no reduction in its bending strength is observed. This implies a disruption in the forming of geopolymer binder, while the WW cellulose structure remains largely intact. A two-fold cause is suggested. First, hydroxide dissolves hemicellulose from untreated WW and releases carboxylic acid, weakening the alkali medium (Cheng-Yong et al., 2023) of the alkali activator. Second, the optimal value of the modulus correlates with its Na_2O concentrations, achieving an equilibrium between alkaline hydrolysis and the absorption rate of soluble Si (Sun et al., 2022). Excess SiO_2 may induce polymerized SiO_4 that suppresses the subsequent geopolymerisation process (Sun et al., 2022).

Generally, increasing GGBFS content in an FA-based geopolymer tends to improve its overall compressive strength (Nath and Sarker, 2014). However, in this study, the inclusion of 10 % or more GGBFS results in a paste with low workability and a shortened setting time. The adhesion between the geopolymer binder and WW fibres is subsequently compromised, especially in samples with untreated WW (sample C4).

Further discussions on the formation of aluminosilicate products will be provided through subsequent FTIR and TG analyses.

The disparity in bending strength between WWGB samples made with treated and untreated WW is more pronounced compared to the differences in their compressive strengths. Generally, WWGB samples incorporating treated WW demonstrate higher bending strengths. This improvement can be attributed to the removal of non-cellulose components, such as wax, and the presence of mercerised cellulose structures in treated WW, which enhance adhesion with the geopolymer gels (Huang et al., 2021; Wimmer et al., 2023). It is worth noting that WWGB samples incorporating treated fibres only exhibit slightly better compressive strength compared to untreated samples. This indicates

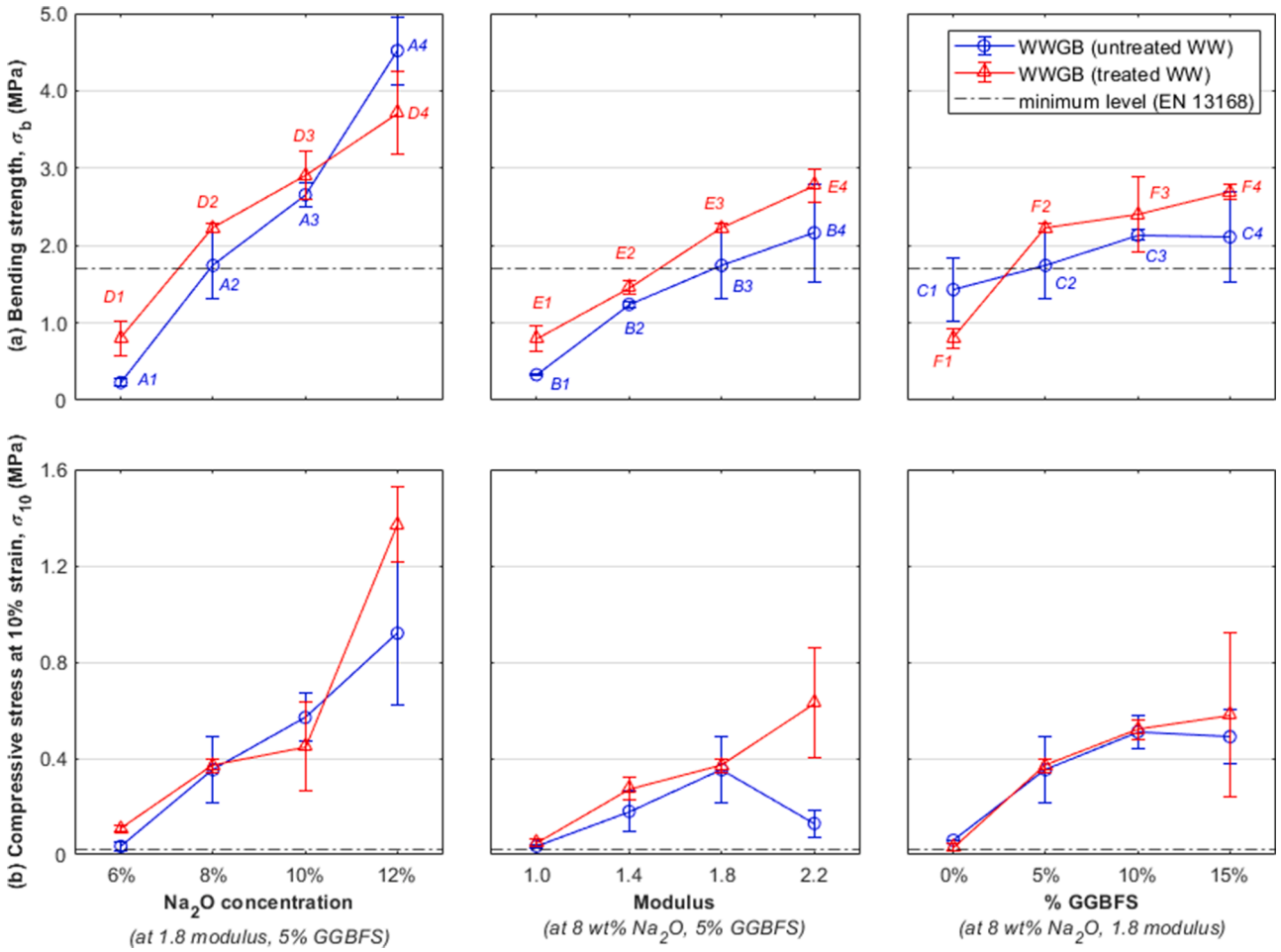


Fig. 2. (a) Bending and (b) compressive strengths of WWGB samples as a function of Na_2O concentration, Modulus, and %GGBFS.

that the geopolymer binder, rather than the fibres, primarily determines the compressive strength, as the formation of aluminosilicate products plays a crucial role in providing interconnecting strength within the composites.

No reduction in bending strength is observed in this study, with alkali activators up to 12 wt.% Na_2O concentration. The structural cellulose largely remains unaffected because alkaline hydrolysis is limited to the surface layer (waxes, oils) and hemicelluloses, either during the fibre pre-treatment or geopolymerisation process. This is aligned with the previous study (Stapper et al., 2021), which suggested that alkalinity is not the main degradation mechanism of natural fibre but other mechanisms such as fibre mineralisation are a bigger influence. It is expected that the long-term performance of the WW will not be significantly compromised by the alkaline pore solution in the geopolymer; however, further study is warranted.

3.2. Insulation properties

The primary objective of WWGB is to enhance thermal insulation in building envelopes. The thermal conductivities of all samples are presented in Fig. 3a. Generally, heat transfer primarily occurs through the interconnected gel network within WWGB. Samples with higher porosity exhibit lower thermal conductivities due to their porous structure, where the internal air voids, including mesopores and macropores, serve as the main insulating component. The results indicate an increase in thermal conductivities with higher Na_2O concentration, modulus and GGBFS content, suggesting a denser gel structure within the samples, which correlates with their mechanical strength. Interestingly, WWGB made of untreated WW demonstrates lower thermal conductivities compared to the treated counterpart. This reveals a trade-off associated with fibre pre-treatment; while it improves mechanical strength, it sacrifices thermal insulation performance due to the denser structure within treated fibres. The median thermal conductivities for all groups range between 70 and 85 mW m^{-1} K^{-1}. These values are comparable to commercially available WWCB products, such as 80 and 90 mW m^{-1} K^{-1} for WWCB with densities of 390 and 420 kg m^{-3}, respectively (MASEA, 2023). Notably, sample A1 exhibits an even lower thermal conductivity below 70 mW m^{-1} K^{-1}; however, its porous structure sacrifices its mechanical strength.

Fig. 3b shows the moisture contents of all WWGB samples conditioned at 80 % relative humidity. Overall, the moisture sorption of the WWGB can be considered low compared to typical WWCB, which has a moisture content of 12 % at 80 % relative humidity (MASEA, 2023), slightly higher than the average of 10 % measured in this study.

From Fig. 3c, the sorption capacities of WWGB are reduced compared to the original WW component under lower relative humidity, with smaller differences at higher relative humidity. Mesopores can be found in WW (Zauer et al., 2014), and when they are not covered by geopolymer, water initially fills up these mesopores at lower relative humidity. In contrast, in the geopolymer paste, there is a combination of macropores and mesopores (Babaee and Castel, 2018), along with the macropore structure in the WWGB, where moisture only fills up in these larger pores at higher relative humidity. Under different geopolymer compositions, a slight increase in moisture content can be observed with higher Na_2O concentration or lower modulus of the alkali activator, while no notable trend is discerned with different GGBFS percentages in

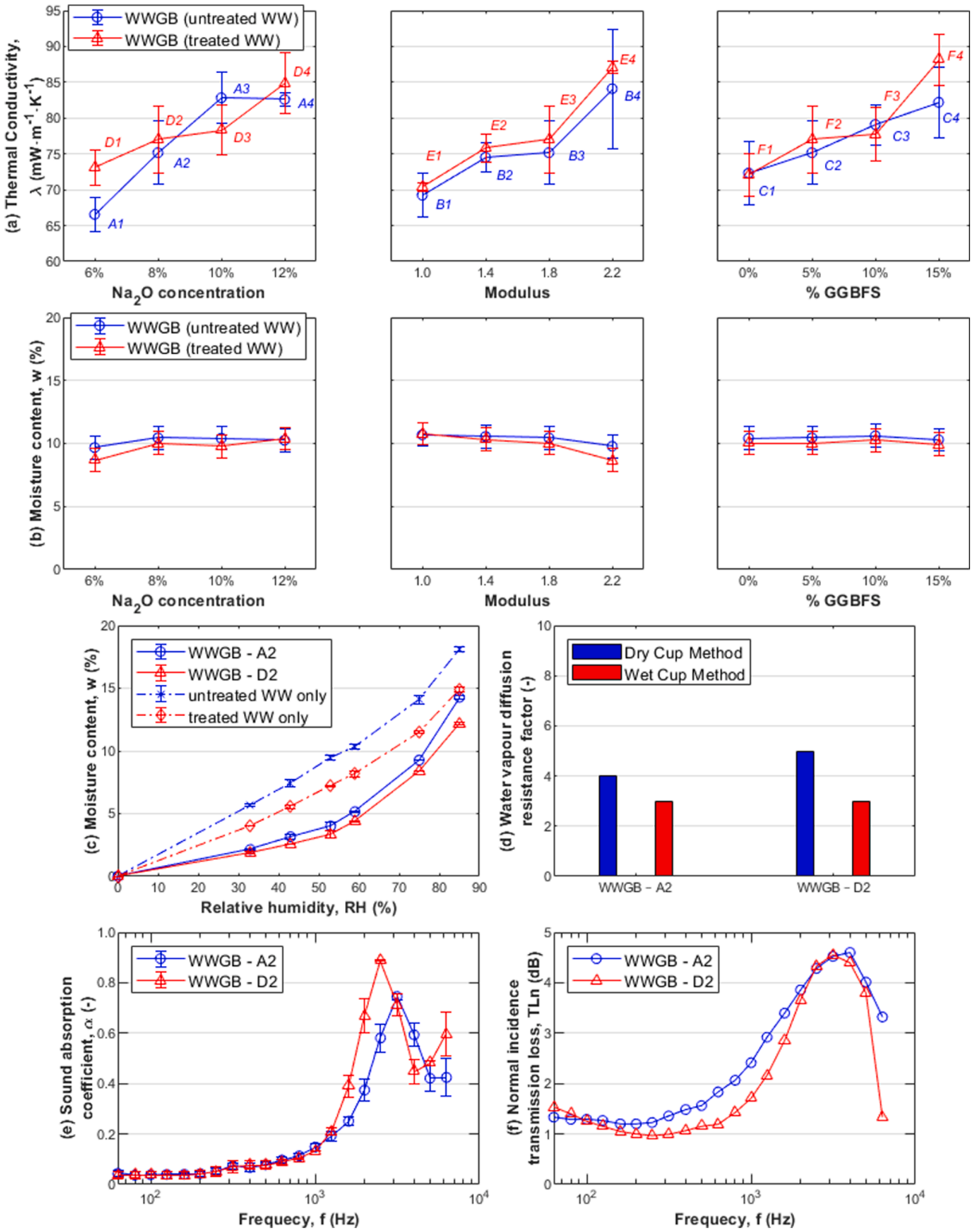


Fig. 3. (a) Thermal conductivities and (b) moisture content at 80 % relative humidity of WWGB samples as a function of Na_2O concentration, Modulus, and %GGBFS. (c) Sorption isotherm, (d) water vapour diffusion resistance factor, (e) sound absorption coefficient, and (f) normal incidence of transmission loss of WWGB samples A2 and D2.

the precursor. This may indicate the creation of a denser network of mesopore structures with higher Na_2O concentration or lower modulus (Luo et al., 2022a,b).

It can also be observed that the WW treated with NaOH solution exhibits lower sorption isotherms compared to the untreated WW. The alkali treatment degraded the cell wall material and reduced the number of hydroxyl groups accessible to water, resulting in increased hydrophobicity (Wimmer et al., 2023; Chandrasekar et al., 2017). Accordingly, the WWGB made from untreated WW (sample A2) has slightly higher sorption isotherms compared to the WWGB with treated WW (sample D2). These consistent trends suggest that the moisture sorption in WWGB is primarily influenced by the fibre component. This observation is further supported by the moisture contents of all WWGB samples conditioned at 80 % relative humidity, where the WWGB with treated WW has lower moisture contents compared to their counterparts.

The water vapour diffusion resistance factors for the samples are found to be between 3 and 5 using the wet cup and dry cup method (Fig. 3d). The low resistance factors indicate that the WWGB has a high moisture diffusivity property. This value is similar to that of typical WWCB, which has a factor of 3 (at a density of 420 kg m^{-3}) (MASEA, 2023).

Fig. 3e shows the normal incidence sound absorption coefficient of Samples A2 and D2. Both samples exhibited a similar trend in sound absorption within the tested frequency range, with the WWGB made from treated WW showing higher absorption coefficients at higher frequencies. Sample D2 shows an absorption peak at 2500 Hz, while sample A2 exhibits a peak at 3150 Hz. These sound absorption patterns are comparable to previous findings on WWCB (Botterman et al., 2018). However, due to the small sample size used in this measurement, the effect of WW treatment is not conclusive. Both samples have a similar sound absorption rating of Class E, with a weighted sound absorption coefficient α_w of 0.15 according to ISO 11654, indicating poor sound absorption (low absorption capability). In comparison, commercial WWCB available in the market achieves an α_w of 0.35 (Class D) when employing a 25 mm thick WWCB attached to a concrete layer (Heraklith, 2023). This suggests that the sound absorption capability of the WWGB samples is relatively close to that of a commercial WWCB.

Both samples, A2 and D2, demonstrate low normal incidence transmission losses below 5 dB, particularly at lower frequencies (Fig. 3f). These suboptimal transmission losses can be attributed to the uneven circumference of the samples caused by their porous structure, leading to flanking transmission during testing. Additionally, the developed WWGB are thin and lightweight, which is not ideal for effective standalone sound insulation (Rindel, 2018).

3.3. Physical properties

The alkali treatment of WW fibres leads to significant changes in surface morphology, as illustrated in Fig. 4b, compared to untreated WW fibres (Fig. 4a). The treated fibres display signs of fibrillation. The reaction between NaOH and the hydroxyl groups present in the wood fibres results in the removal of a portion of lignin, hemicellulose, and wax from the fibres (Chandrasekar et al., 2017). This mercerised cellulose structure enhances interfacial bonding between the fibres and geopolymer gels (Huang et al., 2021; Nguyen and Mangat, 2020) in WWGB. Fig. 4c shows the geopolymer gel on the surface and between the fibres provides the primary binding strength in the WWGB. Unreacted and partially reacted FA particles can also be observed on the WWGB, indicating incomplete reaction of the FA particles during the geopolymerisation process, similar to previous reports (Ismail et al., 2014; Fauzi et al., 2016).

The micro-CT scan of the cross-section of WWGB sample A2 (Fig. 4d) reveals three main components: geopolymer gel (shown in solid grey), pores and voids (shown in solid black) and WW (shown in grey dotted with black). The geopolymer gels, with varying thicknesses, including some exceeding 1 mm, can be observed wrapping around the WW, forming an interconnected network throughout the structure of the WWGB. The original mesopore structure within the WW fibres remains distinguishable. Additionally, new macropores are formed between the coated WW fibres, and mesopores can be observed within the geopolymer gel. Overall, the internal structure exhibits a high level of inhomogeneity. This inhomogeneity highlights the challenges associated with maintaining consistency in the final product at a smaller laboratory scale. However, these challenges could potentially be resolved by scaling up production to the larger scale typically used for WWCB production (Elten, 2006).

In general, the samples exhibit similar ranges of bulk densities, ranging between 350 and 450 kg m^{-3}, depending on the specific design parameters (Fig. 4e). The bulk densities show the greatest increase with an increment in Na_2O concentration, followed by an increase in silica modulus, and were least affected by the percentage of GGBFS in the precursor. These findings align with the general trend observed in mechanical strength and insulation performance; samples characterized by denser structures exhibit both higher thermal conductivity (Fig. 3a) and mechanical strength (Fig. 2).

The open porosities of the fabricated WWGB range from 70 % to 80 % (Fig. 4f). In most cases, samples made with either untreated or treated WW exhibit similar porosity under the same designed parameters. The open porosity shows a clear decreasing trend with an increase in Na_2O concentration owing to the promoted activation; however, no clear trend can be observed when changing the modulus of alkali activators or the percentage of GGBFS in precursors.

3.4. Chemical compositions

The alkali treatment of WW fibres leads to delignification, resulting in the disappearance of characteristic peaks associated with lignin vibrations at 1250 cm^{-1} and 1025 cm^{-1}, which represent the C—O ester vibrations (Collier et al., 2024) (Fig. 5a). Additionally, the peak at 1510 cm^{-1} attributed to aromatic skeletal vibration (Kumar et al., 2022) is no longer observed. These observations align with the fibrillation observed in the treated WW (Fig. 4b). The treated fibres also exhibit a reduced intensity of the band at 3300 cm^{-1}, which is associated with the OH group (Qian et al., 2022). This reduction suggests the disruption of intermolecular hydrogen bonding in cellulose or a decrease in free water content within the treated fibres.

Fig. 5b displays the main peaks at broad bands 1030 cm^{-1} and 900 cm^{-1} for FA and GGBFS respectively. These peaks correspond to the asymmetric stretching vibration modes of Si—O—T bonds (T = tetrahedral Si or Al), exhibiting different glassy networks for FA and GGBFS (Gao et al., 2015a; Criado et al., 2007). Upon alkali activation, the bands for all samples in this study are centred at or around 1000 cm^{-1}, which is typical for an FA-based geopolymer (Fernández-Jiménez and Palomo, 2005). This suggests the formation of a highly polymerized aluminosilicate network as the primary reaction product.

In a blended FA and GGBFS system, the shifting of the main Si—O—T band primarily reflects changes in the polymerization degree of Si—O—T crosslinking. In this study, an increase in the GGBFS ratio in the precursor causes the main Si—O—T band to shift from 1000 cm^{-1} to a lower wavenumber at 980 cm^{-1} (Fig. 5c/iii), indicating a lower polymerization degree due to the presence of a calcium source from GGBFS (Wang et al., 2022). Varying Ca concentrations also impact the formation of geopolymer gel composition, and can lead to the formation of N-A-S-H, C-A-S-H, or intermediate gel (N-C-A-S-H and C-N-A-S-H) (Luo et al., 2023). However, the exact nature of the gels formed are still disputed (Criado et al., 2007; Park et al., 2016; Ismail et al., 2014). C-N-A-S-H type gel is assumed to be predominantly formed in samples with 5 % or higher percentage of GGBFS in the precursor.

Furthermore, the choice of Na_2O percentage also influences the final gel structure. The alkalinity in the environment controls the dissolution of silicate and aluminate from FA and the availability of calcium from

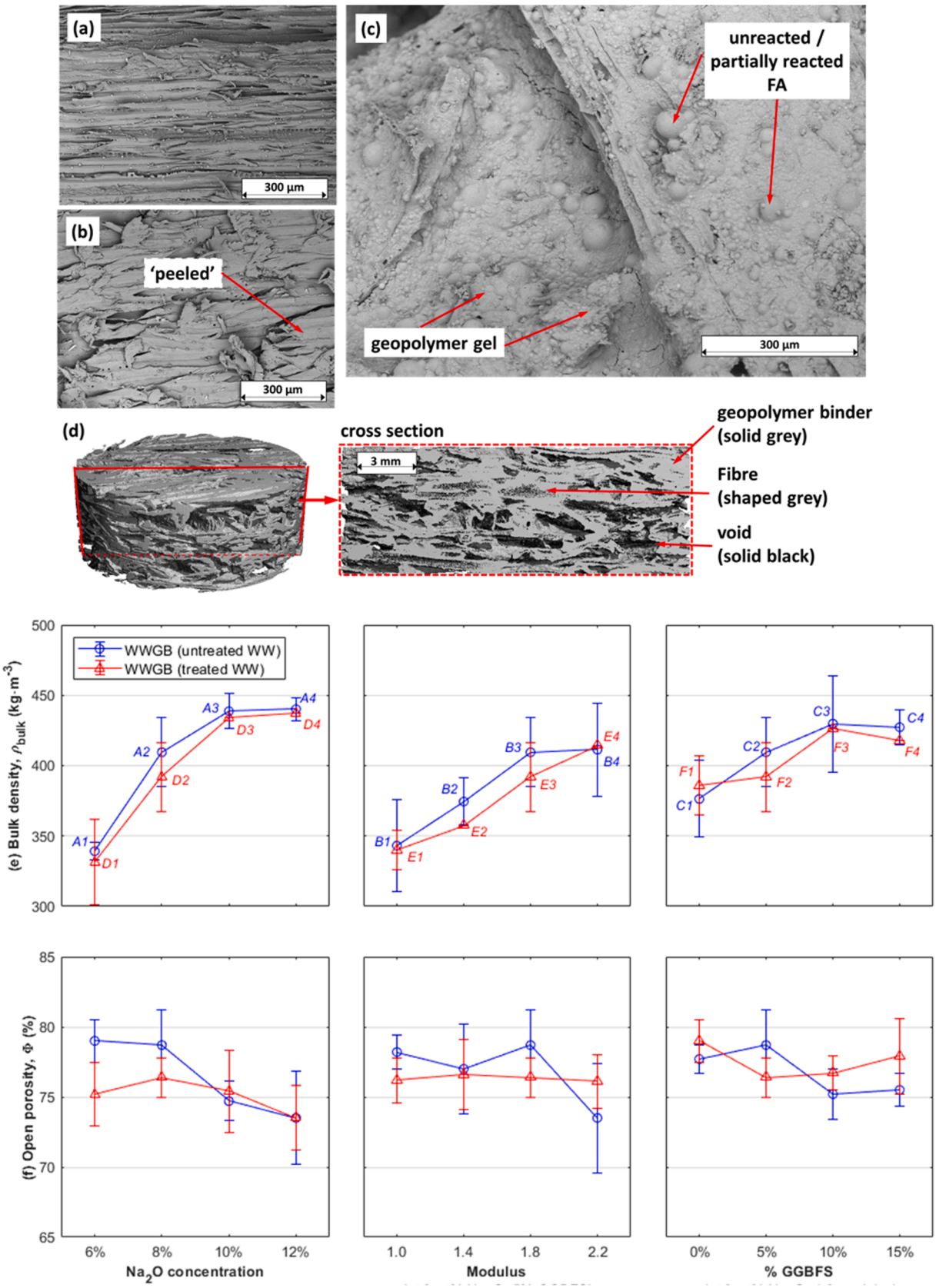


Fig. 4. SEM images of (a) untreated wood wool, (b) treated wood wool, and (c) WWGB. (d) WWGB cross-section from micro-CT scan. (e) Bulk densities and (f) open porosities of WWGB samples as a function of Na_2O concentration, Modulus, and %GGBFS.

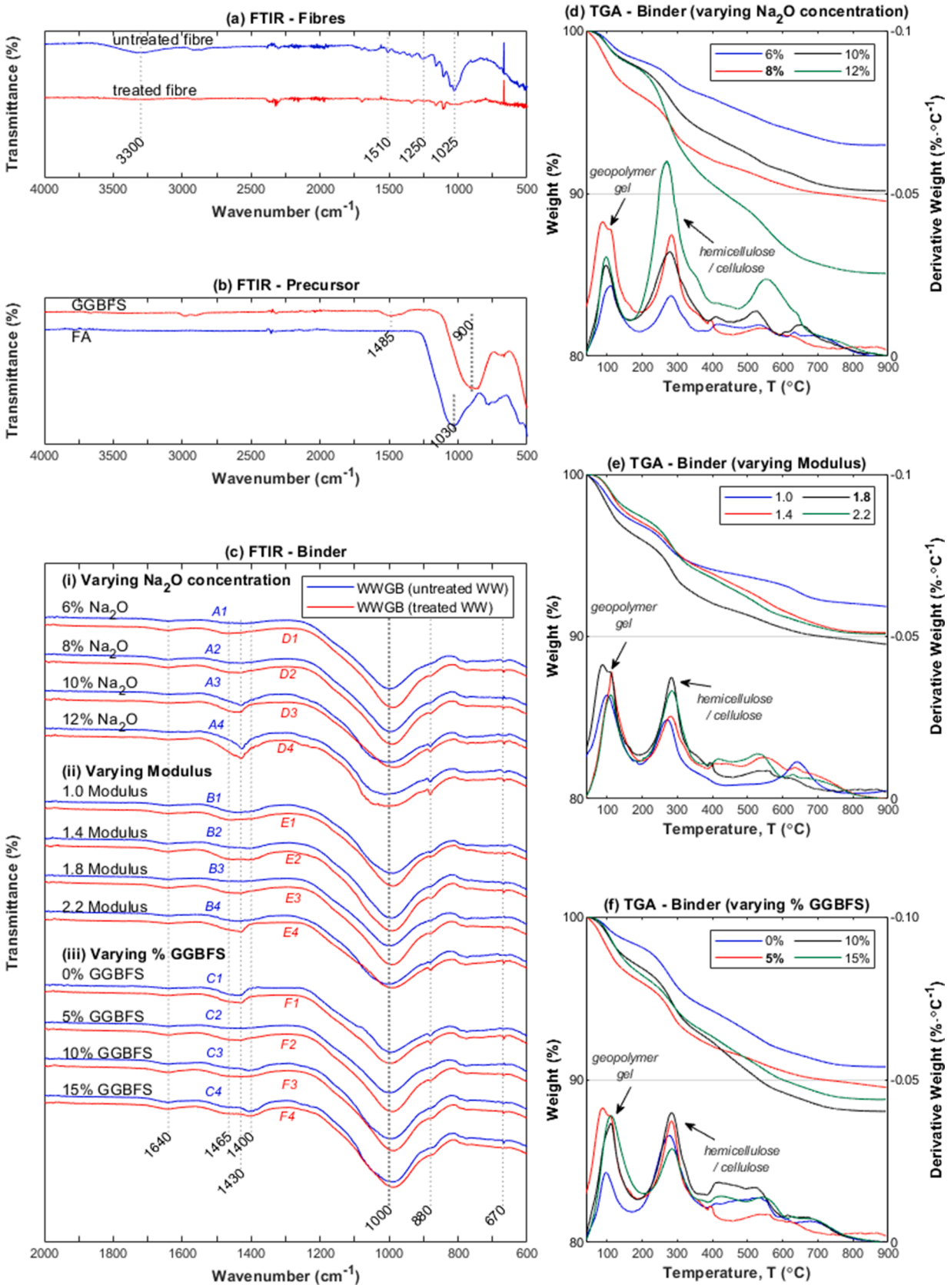


Fig. 5. FTIR results of (a) wood wool fibres, (b) precursors, and (c) binder gel as a function of Na_2O concentrations, modulus, and %GGBFS; TGA and DTG results of WWGB samples (primarily composed of the binder) as a function of (d) Na_2O concentration, (e) Modulus, and (f) %GGBFS.

GGBFS (Ismail et al., 2014). This effect is demonstrated in Fig. 5c/i, where samples with 10 wt.% and 12 wt.% Na_2O concentration exhibit a broader and higher wavenumber shift in the Si—O—T band, indicating variations in the composition/structure of the hybrid C-N-A-S-H gel (Luo et al., 2023). Moreover, a higher modulus increases the availability of silica and raises the pH of the activator, which, in turn, influences the dissolution of precursors and alters the polymerization process of the geopolymer gel (Zhang et al., 2021). This behaviour is evident in Fig. 5c/ii, where the Si—O—T band shifts to a broader and higher wavenumber with a larger modulus, implying a higher polymerization of the final aluminosilicate gel.

The observed shifts in wavenumber correlate with trends in compressive strength (Fig. 2b), illustrating a broader and higher shift associated with the increase in Na_2O concentration compared to variations in modulus and GGBFS content. Additionally, sample B4 manifests a higher peak near 1030 cm^{-1}, indicating the present of unreacted FA, correlating with the observed low compressive strength.

All recipes display a similar small peak at 1640 cm^{-1} (and 3350 cm^{-1}, not presented in Fig. 5c), which can be attributed to the OH groups (Gao et al., 2015a) originating from chemically bound water in the gels.

Carbonation products are detected in both the raw materials and reaction products. For GGBFS, the peak at 1480 cm^{-1} corresponds to the asymmetric stretching mode of the O—C—O bonds, indicating that carbonation had already occurred (Gao et al., 2015b). The reaction products exhibit peaks at 1465 cm^{-1}, 1430 cm^{-1} and 880 cm^{-1}, corresponding to vibrations of O—C—O in carbonates, revealing that carbonation took place during the forming and curing process (Gao et al., 2015a, 2015b). However, in samples with higher GGBFS ratios (10 % and 15 % GGBFS), an additional band at 1400 cm^{-1} is observed, corresponding to O—C—O vibrations from the additional GGBFS content (Ismail et al., 2014).

Fig. 5d–f illustrates the thermogravimetric analysis results of the extracted geopolymer binder powder obtained from WWGB after the thorough removal of fibres. The initial derivative thermogravimetry (DTG) peak is observed in the temperature range of 40 to 200 °C, indicating the release of physically and chemically bound water present in the geopolymer gel structure (Ismail et al., 2014). The second peak, spanning from 200 to 400 °C, corresponds to the decomposition of hemicellulose and cellulose derived from the spruce WW (Grønli et al., 2002) that is mixed within the extracted powder. Additionally, several DTG peaks are observed between 500 and 700 °C, suggesting the presence of carbonates (Zhang et al., 2021). It is worth noting that the variations in mass loss, corresponding to gel decomposition, are more pronounced in the WWGB samples with different Na_2O concentrations compared to adjustments in the silica modulus and GGBFS ratio in the precursor. This observation suggests that the Na_2O concentration plays a more influential role in the formation of the aluminosilicate product, consistent with the observation that the most significant increase in mechanical strength is observed with higher Na_2O concentration.

4. Conclusion

This study demonstrates the substitution of OPC with geopolymer in the production of wood wool geopolymer boards (WWGB). Here are the key findings and characteristics of WWGB:

i. WWGB samples with at least 8 % Na_2O concentration, a modulus of 1.8 or higher, and a GGBFS ratio above 5 % meet both the prescribed minimum compressive strength (20 kPa) and bending strength (1700 kPa) requirements. Mechanical strength is primarily influenced by the Na_2O concentration, followed by the GGBFS percentage and the modulus, as supported by FTIR and TG analyses indicating an increased reaction degree.
ii. The present of natural fibres impacts the optimal value of Na_2O concentration and modulus for compressive strength. Treated fibres are recommended when applying a higher modulus to avoid hindrance in the geopolymerisation process. Furthermore, WWGB made from treated wood wool exhibits higher bending strengths.
iii. Median thermal conductivities of WWGB fall within the range between 70 and 85 mW m^{-1} K^{-1}. WWGB made from treated wood wool demonstrates slightly higher thermal conductivities compared to the untreated counterpart, revealing a trade-off associated with fibre pre-treatment.
iv. Moisture sorption in WWGB is primarily influenced by the fibre component, with untreated WWGB exhibiting slightly higher sorption isotherms. WWGB show low moisture sorption and water vapour diffusion resistance.
v. Bulk densities of WWGB range from 350 to 450 kg m^{-3}, and open porosities range from 70 % to 80 %. These properties are primarily influenced by the Na_2O concentration and silica modulus, while the GGBFS percentage has minimal impact.

In conclusion, this study underscores geopolymer as a viable alternative to OPC in WWGB production, providing insights into manufacturing methods, design parameters and performances. Opportunities for improvements include enhancing acoustic performance. Further investigations should assess fire resistance and durability, explore alternative natural fibre materials, conduct a comprehensive life cycle assessment, and optimize curing, drying, and alkali activator for lower embodied carbon.

Spotlights

1. Substitute ordinary Portland cement (OPC) with geopolymers in the production of wood wool geopolymer boards (WWGB).
2. The WWGB are lightweight, meet the strength requirements, low thermal conductivity and low moisture sorption.
3. The WWGB offer comparable properties to conventional wood wool cement boards while reducing embodied carbon.
4. Industrial waste fly ash and ground granulated blast furnace slag are utilized as raw material in the board-making.
5. Replacement of wood wool with other natural by-product fibres in the making of insulation boards.

CRediT authorship contribution statement

C.H. Koh: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft. **Y. Luo:** Formal analysis, Methodology, Writing – review & editing. **F. Gauvin:** Supervision, Writing – review & editing. **K. Schollbach:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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